

Database Information

This database includes the de-identified EEG data from 62 healthy individuals who participated in a brain-computer interface (BCI) study. All subjects underwent 7-11 sessions of BCI training which involves controlling a computer cursor to move in one-dimensional and two-dimensional spaces using subject's "intent". EEG data were recorded with 62 electrodes. In addition to the EEG data, behavioral data including the online success rate of BCI cursor control are also included.

This dataset was collected under support from the National Institutes of Health via grants AT009263, EB021027, NS096761, MH114233, RF1MH to Dr. Bin He.

Correspondence about the dataset: Dr. Bin He, Carnegie Mellon University, Department of Biomedical Engineering, Pittsburgh, PA 15213. E-mail: bhe1@andrew.cmu.edu

This dataset has been used and analyzed to study the learning of BCI control and the effects of mind-body awareness training on this process. The results are reported in: Stieger et al, "Mindfulness Improves Brain Computer Interface Performance by Increasing Control over Neural Activity in the Alpha Band," *Cerebral Cortex*, 2020 (<https://doi.org/10.1093/cercor/bhaa234>). Please cite this paper if you use any data included in this dataset.

EEG data organization for BCI tasks

This dataset consists of 598 data files, each of which contains the complete data record of one BCI training session. Each file holds approximately 60 minutes of EEG data recorded during the 3 BCI tasks mentioned above and comprises 450 trials of online BCI control. Each datafile is from one subject and one session and is identified by their subject number and BCI session number.

All data files are shared in .mat format and contain MATLAB-readable records of the raw EEG data and the recording session's metadata described below. The data in each file are represented as an instance of a Matlab structure named "BCI" having the following key fields "data," "time," "positionx," "positiony," "SRATE," "TrialData," "metadata," and "chaninfo". SX_Session_Y.mat is the filename for subject X's data record of their Yth BCI training session. Participants are numbered 1 through 62 and sessions are numbered 1 (i.e., the baseline BCI session that occurs 8 weeks prior to the main BCI training) through 11 (or 7 if the participant only completed 6 post intervention training sessions).

The fields of the structure "BCI" comprising the data record of each file are as follows. The main field is "data," where the EEG traces of each trial of the session are stored. Since each trial varies in length, the "time" data field contains a vector of the trial time (in ms) relative to target presentation. The two position fields, "positionx" and "positiony," contain a vector documenting the cursor position on the screen for each trial. "SRATE" is a constant representing the sampling rate of the EEG recording, which was always 1000Hz in these experiments. "TrialData" is a structure containing all the relevant descriptive information about a given trial. The field "metadata" contains participant specific demographic information. Finally, "chaninfo" is a data structure containing relevant information about individual EEG channels.

The "data" field contains the recording session's EEG data in the format of a 1 x nTrials cell array, where each entry in the cell array is a 2D Matlab array of size nChannels x nTime. The number of trials per session, nTrials, is nearly always 450. Each row of a trial's 2D data matrix is the time-series of voltage measurements (in μV) from a single EEG input lead such as C3 or C4. The "time" field contains the time data in the format of a 1 x nTrials cell array, where each entry is a 1 x nTime vector of the time index of the trial referenced to target presentation. Each trial begins with a 2s inter-trial interval (index 1, $t = -2000\text{ms}$), followed by target presentation for 2s (index 2001, $t = 0$). Feedback begins after target presentation (index 4001, $t = 2000\text{ms}$) and continues until the end of the trial. The end of the trial can be found in the "resultind" subfield of "TrialData" (see below). A post-trial interval of 1s (1000 samples) follows target selection or the timeout signal.

While the length of the trial may vary, `nTime` (the number of time samples of the trial), the number of channels, `nChannels`, is always 62. The ordering of the channels can be found in the `"chaninfo"` structure's subfield `"label"`, which lists the channel names of the 62 electrodes. If individual electrode positions were recorded, the subfield `"positionsrecorded"` will be set to 1 and the `"electrodes," "fiducials,"` and `"shape"` subfields will be populated according to this information. `"Electrodes"` contains the electrode labels and their X, Y, and Z coordinates. The information in the `"fiducials"` subfield contains the X, Y, and Z locations of the nasion as well as the left and right preauricular points. The shape of the face was recorded by sweeping from ear to ear under the chin and then drawing arcs horizontally across the brow and vertically from the forehead to chin. This information is contained in the `"shape"` subfield. If the session specific individual electrode positions are not available, `"positionsrecorded"` will be set to zero, the `"fiducials"` and `"shape"` subfields will be empty, and the `"electrodes"` subfield will include generic electrode locations for the 10-10 system positioning of the Neuroscan Quik-Cap. Finally, the subfield `"noisechan"` identifies which channels were found to be particularly noisy during the BCI session and can be used for easy exclusion of this data or channel interpolation.

The position subfields of the `"BCI"` structure, `"positionx"` and `"positiony,"` provide the cursor positions throughout the trial in the format of a 1 x `nTrials` cell array. Each entry of this cell array is a 1 x `nTime` vector of the horizontal or vertical position of the cursor throughout the trial. Positions are provided by BCI2000 starting with feedback control (index 4001) and continue through the end of the trial (`TrialData(trial).resultind`). The inter-trial intervals before and after feedback are padded with NaNs. The positional information ranges from 0 (left/bottom edge of the screen) to 1 (right/top edge of the screen), with the center of the screen at 0.5. During 1D tasks (LR/UD), the orthogonal position remains constant (e.g., in the LR task, `positiony` = 0.5 throughout the trial).

The `"TrialData"` structure subfield of `"BCI"` provides valuable information for analysis and supervised machine learning (detailed in Table 2). This structure contains the subfields `"tasknumber," "runnumber," "trialnumber," "targetnumber," "triallength," "targethitnumber," "resultind," "result," "forcedresult,"` and `"artifact."` `Tasknumber` is used to identify the individual BCI tasks (1 = 'LR', 2 = 'UD', 3 = '2D'). `Targetnumber` is used to identify which target was presented to the participants (1 = right, 2 = left, 3 = up, 4 = down). `"Triallength"` is the length of the feedback control period in seconds, with the maximum length of 6.04s occurring in timeout trials. `"Resultind"` is the time index that indicates when the feedback ended (when a target was selected or the trial was considered a timeout). To easily extract the feedback portion of the EEG data, you simply define an iterator variable (e.g., `trial = 5`) and index into the data subfield [e.g., `trial_feedback = BCI.data{trial}{4001:BCI.TrialData(trial).resultind}`]. `"Targethitnumber"` identifies which target was selected during feedback and takes the same values as `"targetnumber"` mentioned above. These values will be identical to `"targetnumber"` when the trial is a hit, different when the trial is a miss, and NaN when a trial is a timeout. `"Result"` is a label for the outcome of the trial which takes values of 1 when the correct target was selected, 0 when an incorrect target is selected and NaN if the trial was labeled as a timeout. `"Forcedresult"` takes the same values as before, however in timeout trials, the final cursor position is used to select the target closest to the cursor thereby imposing the BCI system's best guess onto timeout trials. Finally the `"artifact"` subfield of the `"TrialData"` structure is a bool indicating whether an artifact was present in the EEG recording.

Lastly, the `"metadata"` field of the `"BCI"` data structure includes participant and session level demographic information (detailed in Table 3). `"Metadata"` contains information related to the participant as an individual, factors that could influence motor imagery, information regarding the mindfulness intervention, and individual session factors. Participant demographic information includes `"age"` (in years), `"gender"` ('M' for male, 'F' for female, NaN if not available), and `"handedness"` ('R' if right handed, 'L' if left handed, NaN if not available). Motor imagery factors include `"instrument"` ("Do you play a musical instrument?"), `"athlete"` (Do you consider yourself to be an athlete?), `"handsport"` ("Do you play a hand-based sport, for example basketball?"), and `"hobby"` ("Do you have a hobby that requires fine motor

movements of your hands, for example knitting?”). These factors are reported as ‘Y’ for yes, ‘N’ for no, ‘U’ for used to, and NaN if not available. Subfields for the mindfulness intervention include “MBSRsubject” (Did the participant attend the MBSR intervention? 1 = yes, 0 = no) and “meditationpractice” (the hours of meditation practice recorded outside the intervention; NaN if participant was a member of the control group). Finally, session factors included “date” (i.e., the date of the session; yearmonthday), “day” (i.e., the day of the week, starting with 1 for Monday), and “time” (the hour that the session began in military time).